

Eric Gan

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EDUCATION

UC San Diego - 3.87 GPA

Bachelor of Arts in Data Science

La Jolla, CA

Sep. 2023 – Jun 2027

Relevant Coursework: AI: Search and Reasoning (CSE 150B), Genetics (BICD 100), Ecosystems and Global Change (BIEB 174), Introductory Biology Lab (BILD 4)

EXPERIENCE

Undergraduate Research Assistant

Knight Lab, UC San Diego

Feb. 2025 – June 2025

San Diego, CA

- Curated a global gut microbiome meta-dataset by systematically scraping literature from Google Scholar, extracting diverse cohort data, and standardizing metadata fields to ensure cross-study consistency
- Aggregated and visualized large-scale metadata in QIIME 2, analyzing taxonomic distributions and microbial community patterns across distinct geographic populations

Food Service Assistant Manager

Philmont Scout Ranch

June 2024 – August 2024, June 2025 – August 2025

Cimarron, NM

- Supervised food service operations serving up to 300 people per meal
- Created Standard of Operating Procedures to improve operational efficiency and consistency
- Trained and led a team of 7 to ensure safe and efficient service

President

333Jugglers, UC San Diego

December 2024 – Present

San Diego, CA

- Co-founded a recreational juggling club at UCSD with 110 members
- Reserved rooms and secured quarterly club funding
- Coordinated the design and production of club promotional flyers and custom merchandise

PROJECTS

Formula 1 DNF Prediction | *Python, LightGBM, XGBoost, Optuna, Hugging Face Spaces* | [Live Demo](#) June 2026

- Trained and evaluated Logistic Regression, Random Forest, XGBoost, and LightGBM models under significant class imbalance and temporal distribution shifts
- Optimized LightGBM using Optuna Bayesian hyperparameter search over 30 trials, achieving a 2.6x improvement in F1-score and 5x in Recall compared to baselines
- Built and deployed an interactive web application on Hugging Face Spaces for real-time race outcome predictions

1D Convolution Visualization Playground | *React* | [Live Demo](#)

Mar 2026

- Built an interactive web application to visualize discrete convolution with step-by-step kernel sliding
- Designed interactive controls for signals and kernels to improve intuition for linear systems

SoCalGuessr Image Classification | *PyTorch, MobileNetV3, Computer Vision*

Mar 2026

- Built and evaluated a 6-class image classification model using MobileNetV3-Small
- Applied transfer learning by modifying final layers for custom classification
- Achieved ~84% validation accuracy over 20 epochs
- Optimized lightweight architecture (~1.5M parameters) for efficiency

Gait Pattern Analyzer | *Python, D3, JavaScript, HTML, CSS*

June 2025

- Developed interactive visualizations to analyze gait data from neurological datasets
- Used Python for data processing and D3.js for dynamic comparisons
- Designed UI for exploring patient-level gait differences

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS, R

Frameworks & Libraries: React, PyTorch, scikit-learn, pandas, NumPy, Matplotlib, D3.js

Tools & Technologies: Git, Bash, PySpark, Dask, QIIME 2, VS Code, IntelliJ, Cursor, Antigravity